Homework 01

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1.10

suppose x and z are continuous random variables, and the density function of x is $f(x)$ and the density function of z is $g(z)$. Since x and z are independent, the joint density function of x and z is $f(x)g(z)$. Then we have

$$\begin{aligned} E(x+z) &= \int\int (x+z)f(x)g(z)dxdz = \int\int xf(x)g(z)dxdz + \int\int zf(x)g(z)dxdz \\ &= \int xf(x)dx \int g(z)dz + \int f(x)dx \int zg(z)dz = E(x) + E(z) \end{aligned}$$

$$E(xz) = \int\int xzf(x)g(z)dxdz = \int xf(x)dx \int zg(z)dz = E(x)E(z)$$

Hence, we have

$Var(x+z) = E((x+z)^2) - E^2(x+z) = E(x^2) + 2E(x)E(z) + E(z^2) - E^2(x) - 2E(x)E(z) - E^2(z) = Var(x) + Var(z)$

Remark. Actually,$E(x+z) = E(x) + E(z)$ is trus for any random variables x and z, so$E(x^2 + 2xz + z^z) = E(x^2) + 2E(x)E(z) + E(z^2)$ is also true for any random variables x and z.

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(1.55)

$$\frac{d}{d\mu} ln(p|\mu, \sigma^2) = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu) = 0$$

$$N\mu = \sum_{i=1}^{N} x_i$$

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$$

(1.56)

$$\frac{d}{d\sigma^2} ln(p|\mu, \sigma^2) = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{N} (x_i - \mu)^2 = 0$$

$$\sigma^2 = \frac{1}{N} \sum_{n=1}^{N} (x_n - \mu_{ML})^2$$

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if n=m, by (1.50)

$$E(x_n, x_m) = E(x^2) = \mu^2 + \sigma^2 = E(x^2) = \mu^2 + I_{n\,m}\sigma^2$$

if $n \neq m$, by (1.49) and (1-10)

$$E(x_n, x_m) = E(x_n)E(x_m) = \mu^2 = E(x^2) + I_{n\,m}\sigma^2$$

Proof of (1.57),

$$E(\mu_{ML}) = \frac{1}{N}E(\sum_{i=1}^{N} x_i) = \frac{1}{N}\sum_{i=1}^{N} E(x_i) = \frac{1}{N}\sum_{i=1}^{N} \mu = \mu$$

Proof of (1.58),

$$E(\mu_{ML}^2) = \frac{1}{N^2}E(\sum_{n=1}^{N} x_n)^2 = \frac{1}{N^2}E(\sum_{n=1}^{N} x_n^2 + \sum_{n=1}^{N}\sum_{m\neq n} x_n x_m) = \frac{1}{N^2}(N(\mu^2+\sigma^2)+N(N-1)\mu^2) = \mu^2+\frac{\sigma^2}{N}$$

$$\begin{aligned}
E(\sigma_{ML}^2) &= E(\frac{1}{N}\sum_{n=1}^{N}(x_n-\mu_{ML})^2) = \frac{1}{N}\sum_{n=1}^{N} E((x_n-\mu_{ML})^2) \\
&= \frac{1}{N}\sum_{n=1}^{N}[E(x_n^2) - \frac{1}{N}E(2x_n(\sum_{i=1,i\neq n}^{N} x_i) - \frac{2}{N}E(x_n^2) + E(\mu_{ML}^2))] \\
&= \frac{1}{N}\{(N-2)(\mu^2+\sigma^2) - 2(N-1)\mu^2 + N\mu^2 + N\frac{\sigma^2}{N}\} \\
&= \frac{N-1}{N}\sigma^2
\end{aligned}$$